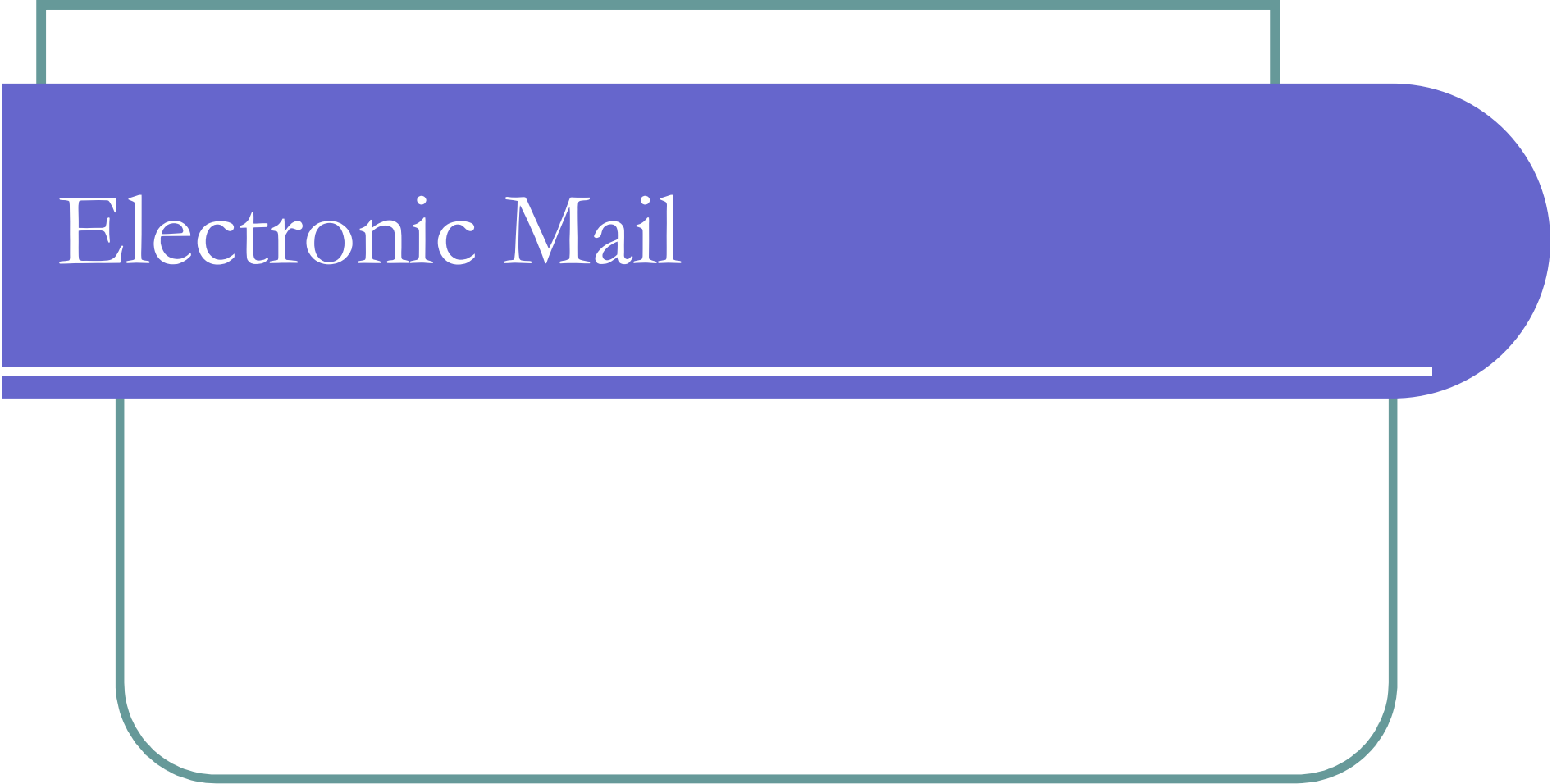




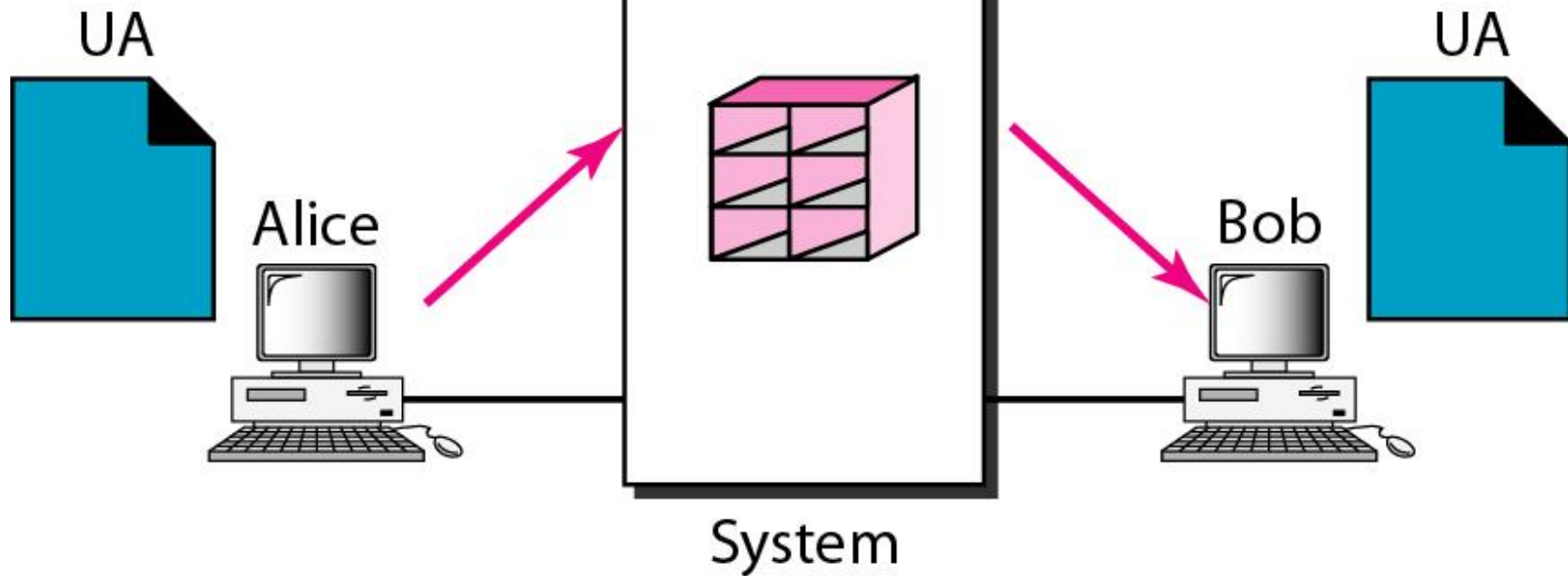
Electronic Mail

Introduction

- One of the most popular Internet services is electronic mail (e-mail).
- The designers of the Internet probably never imagined the popularity of this application program.
- Its architecture consists of several components
 - User Agent
 - Message Transfer Agent: SMTP (Send)
 - Message Access Agent: POP and IMAP (Receive)

First scenario in electronic mail

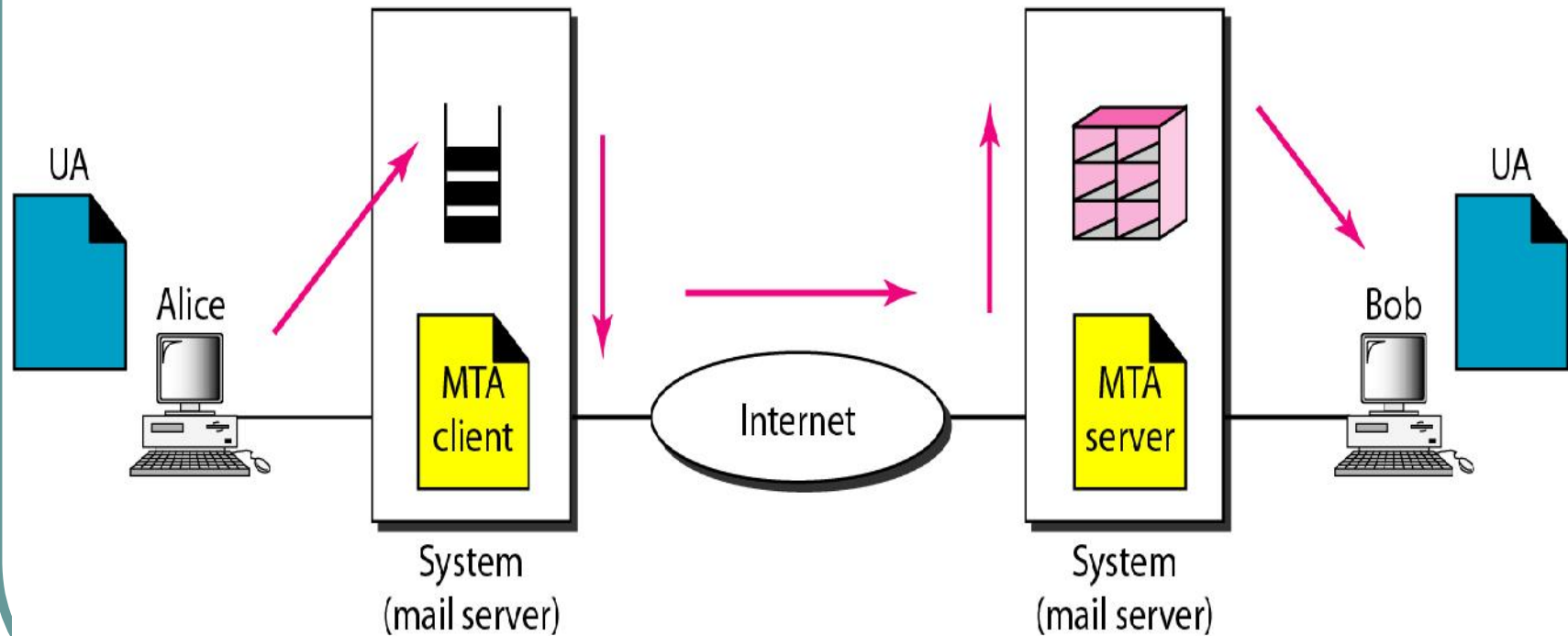
UA: user agent



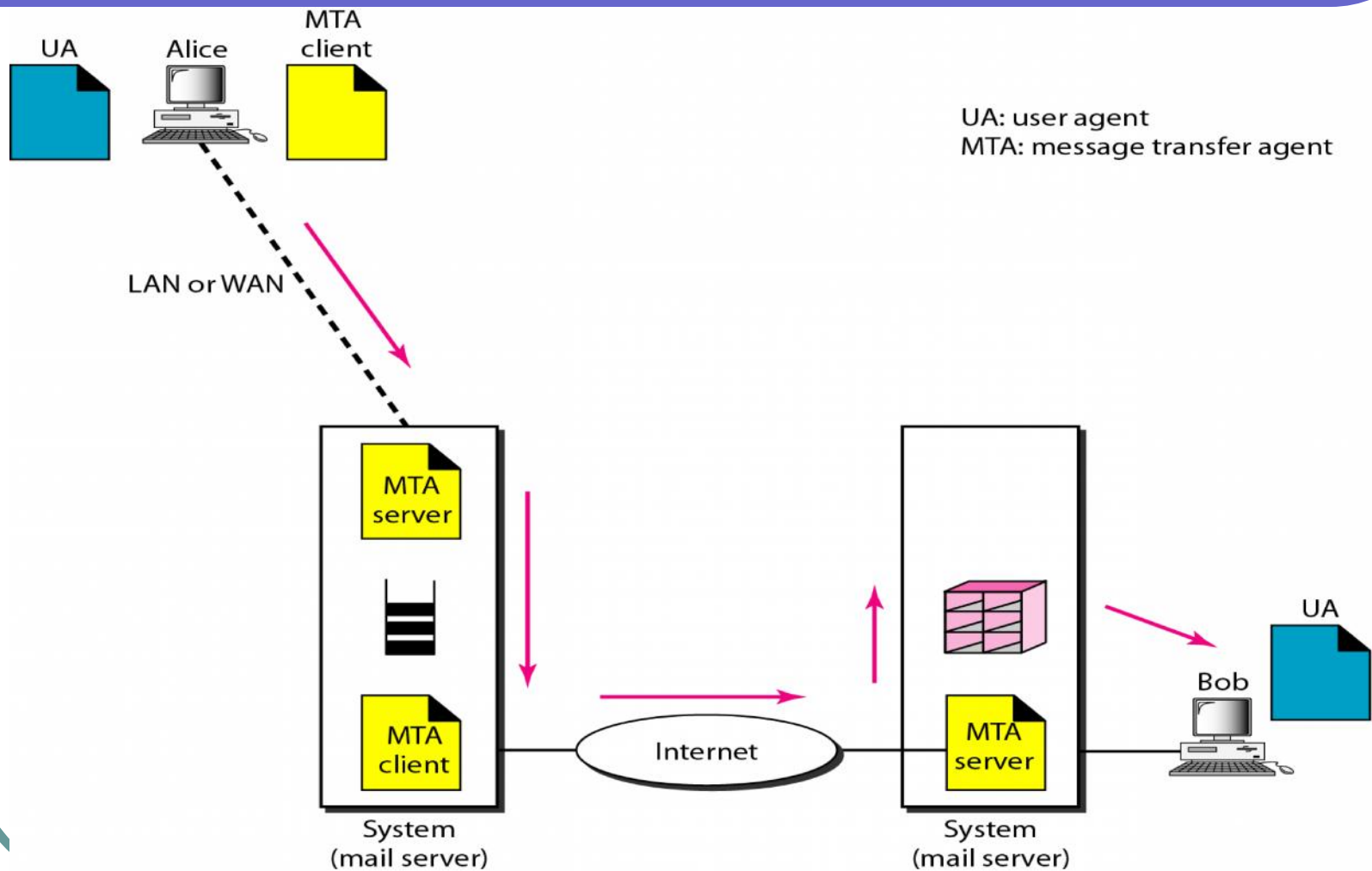
Second scenario in electronic mail

UA: user agent

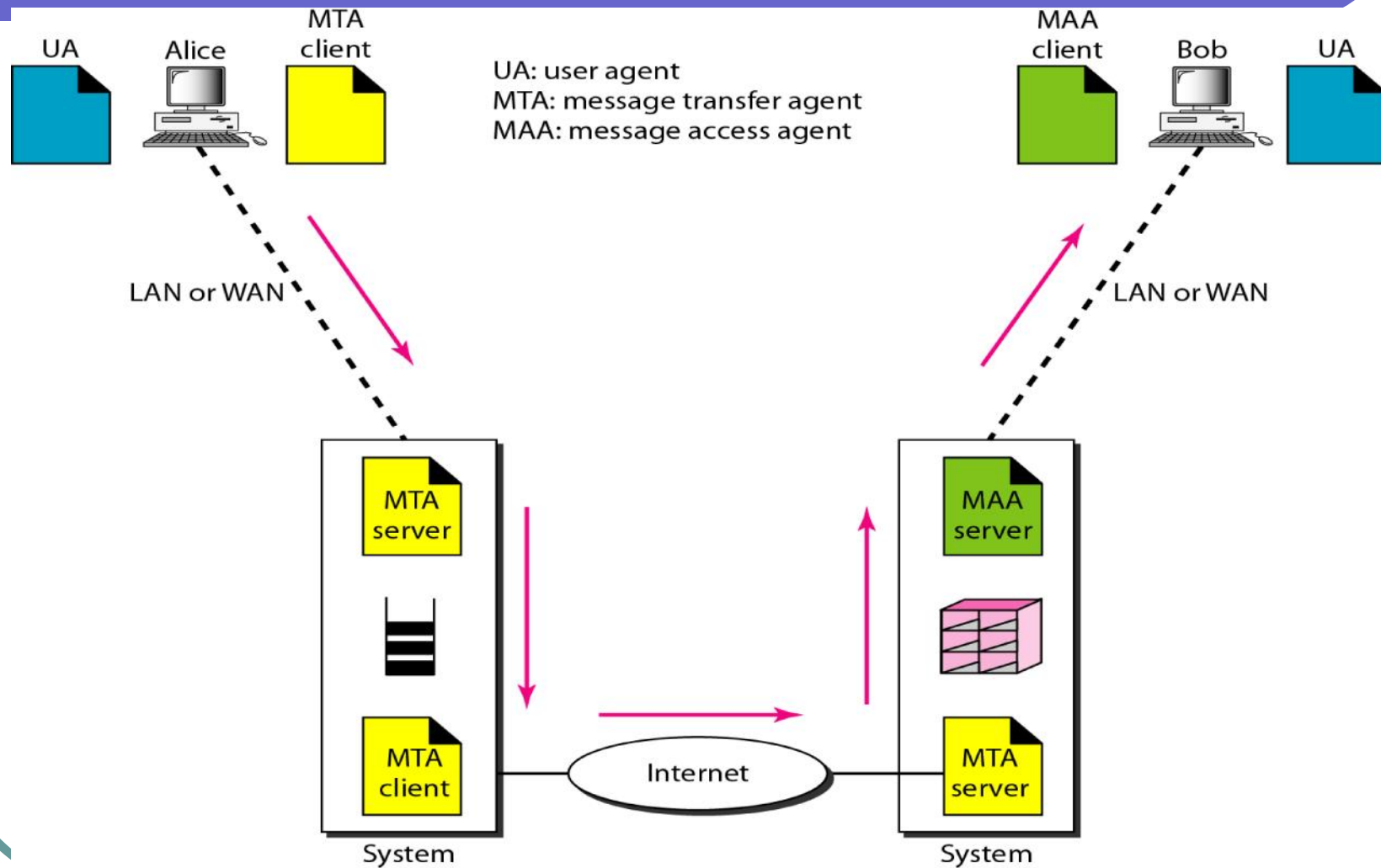
MTA: message transfer agent



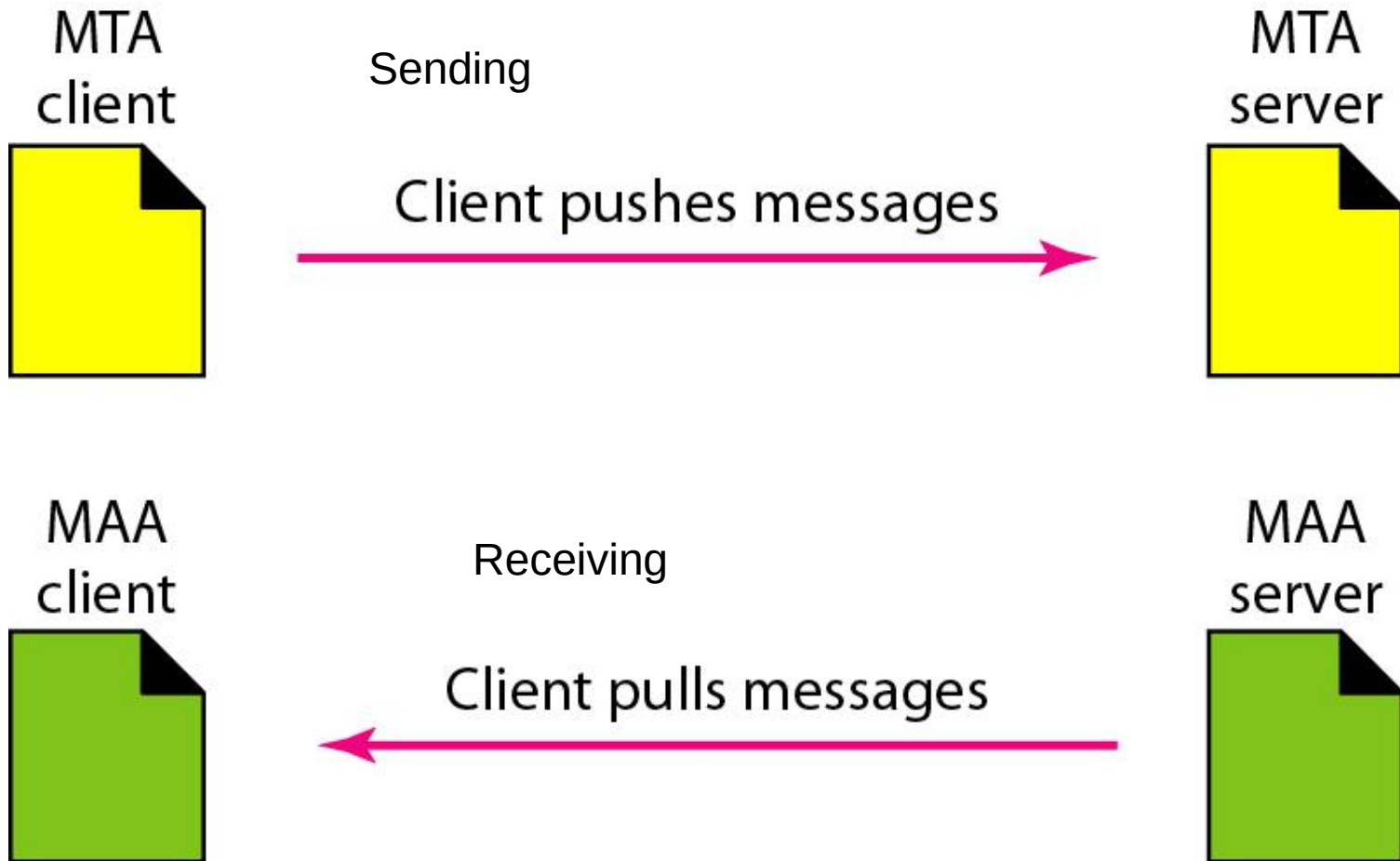
Third scenario in electronic mail



Fourth scenario in electronic mail

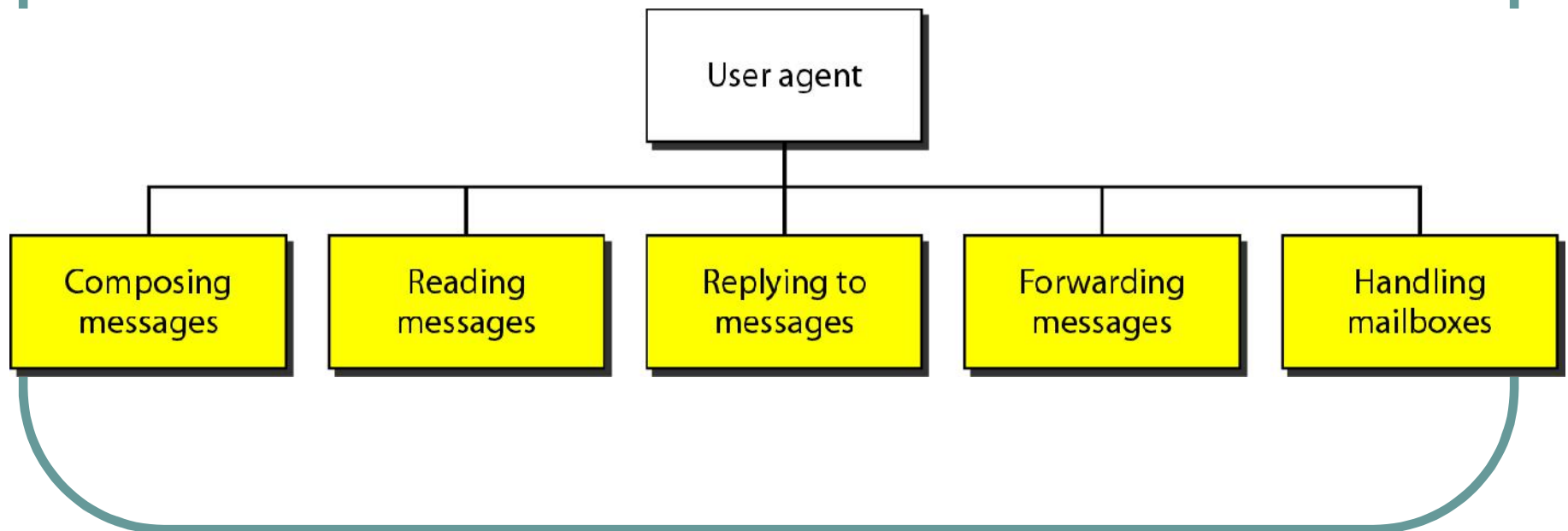


Push versus pull in electronic email



User Agent

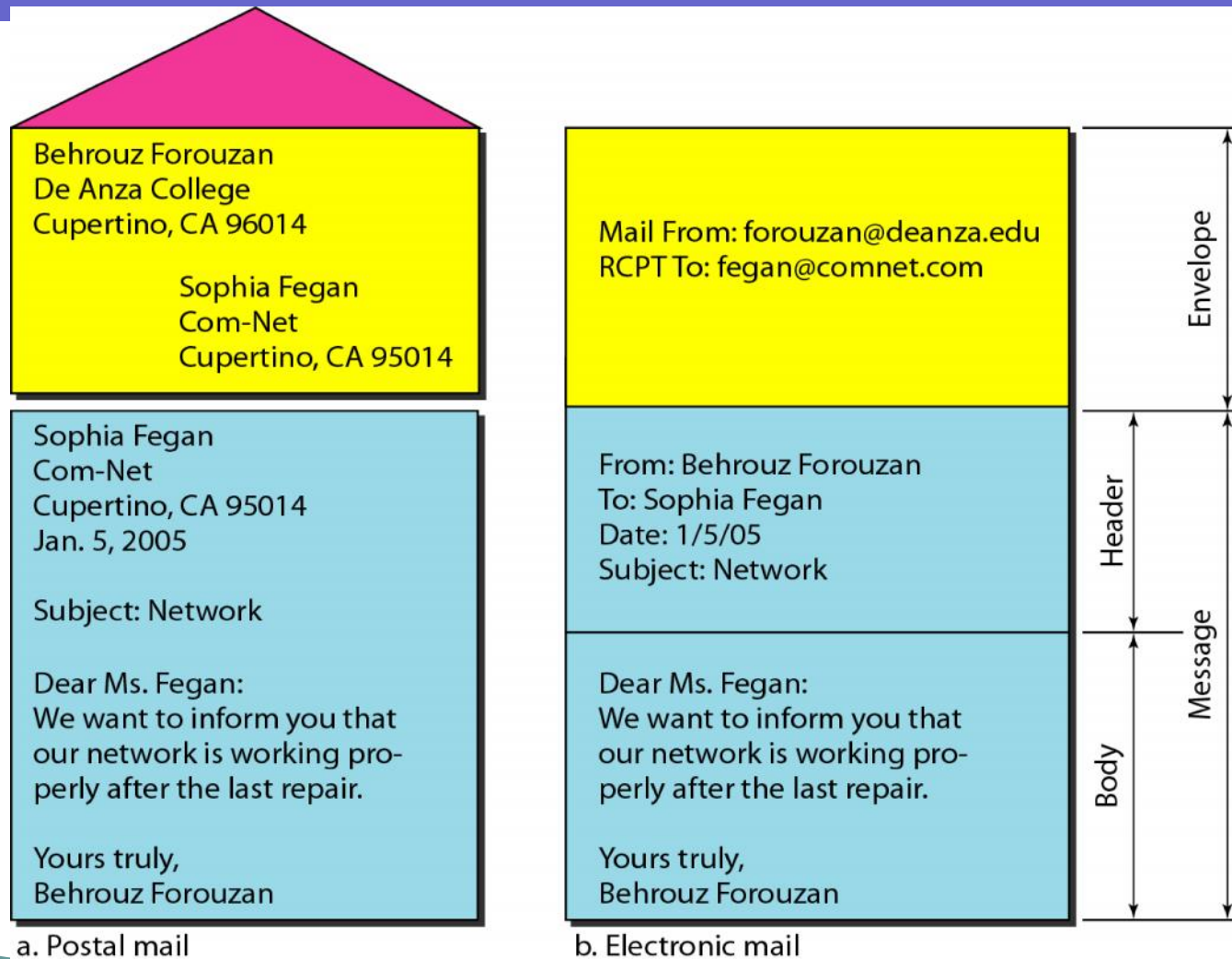
- Provides service to the user to make the process of sending and receiving message easier.
- Services provided by the user agent are:



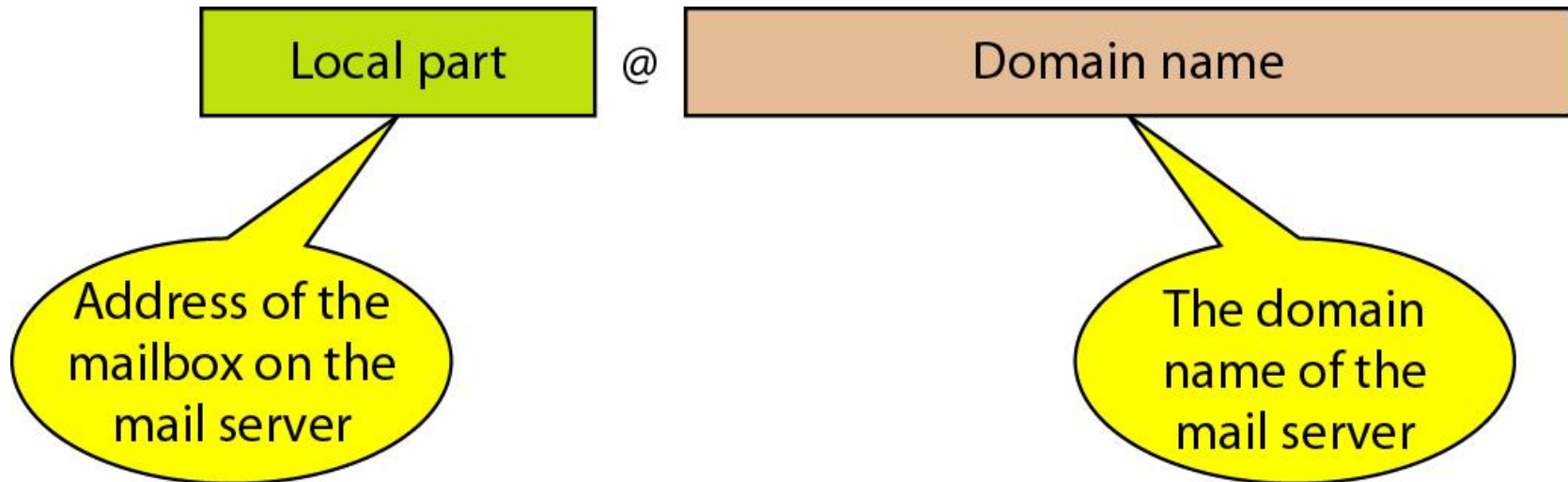
User Agent Types

- Command Driven
 - mail, pine (a Program for Internet News and Email) , and elm
- GUI Based
 - Outlook, and Netscape.

Format of an e-mail



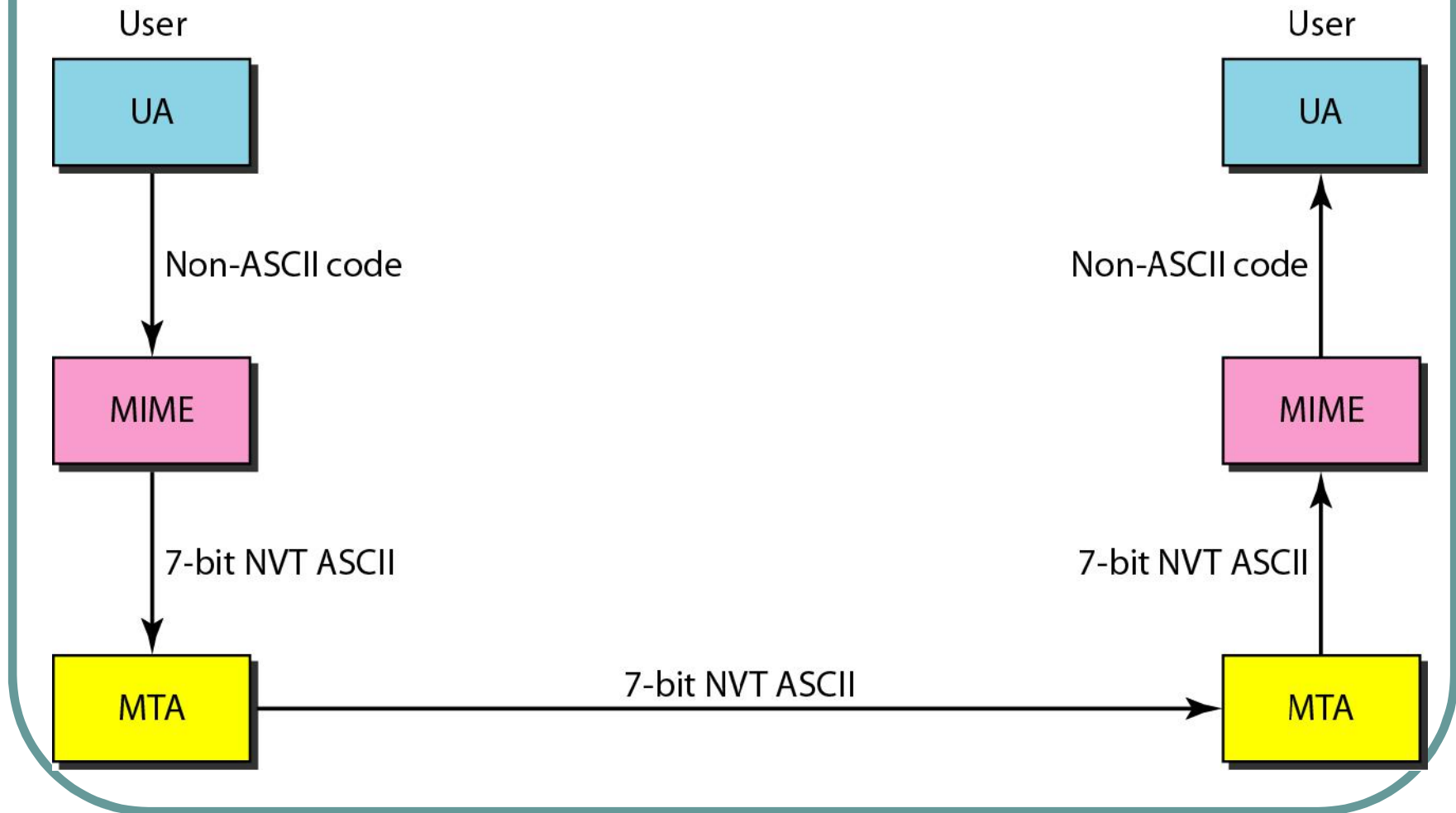
E-mail address



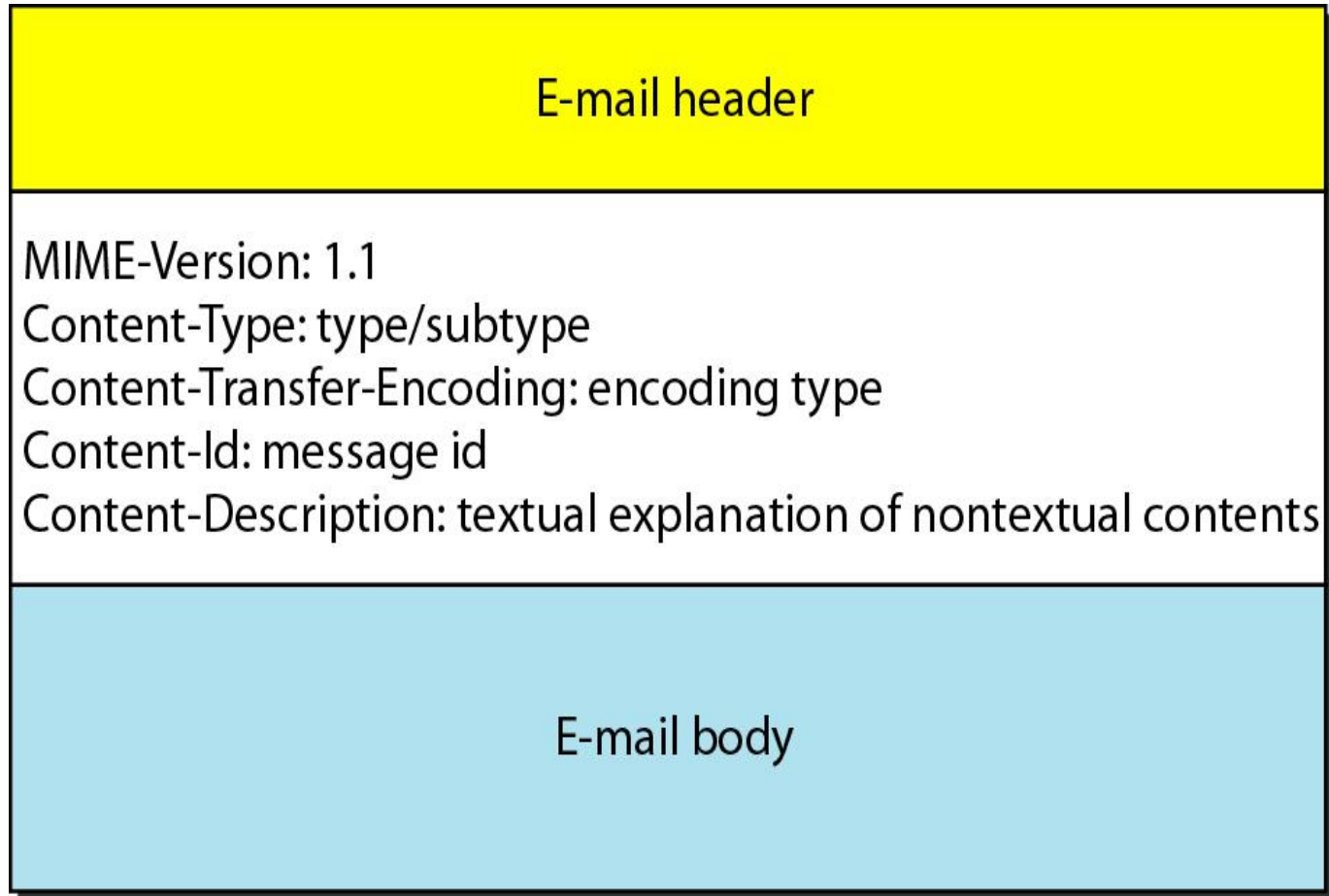
MIME- Multipurpose Internet Mail Extensions

- Allows non ASCII data to be sent through e-mail.
- Set of software functions that transforms non ASCII data to NVT-ASCII and vice versa.
- NVT – Network Virtual Terminal

MIME



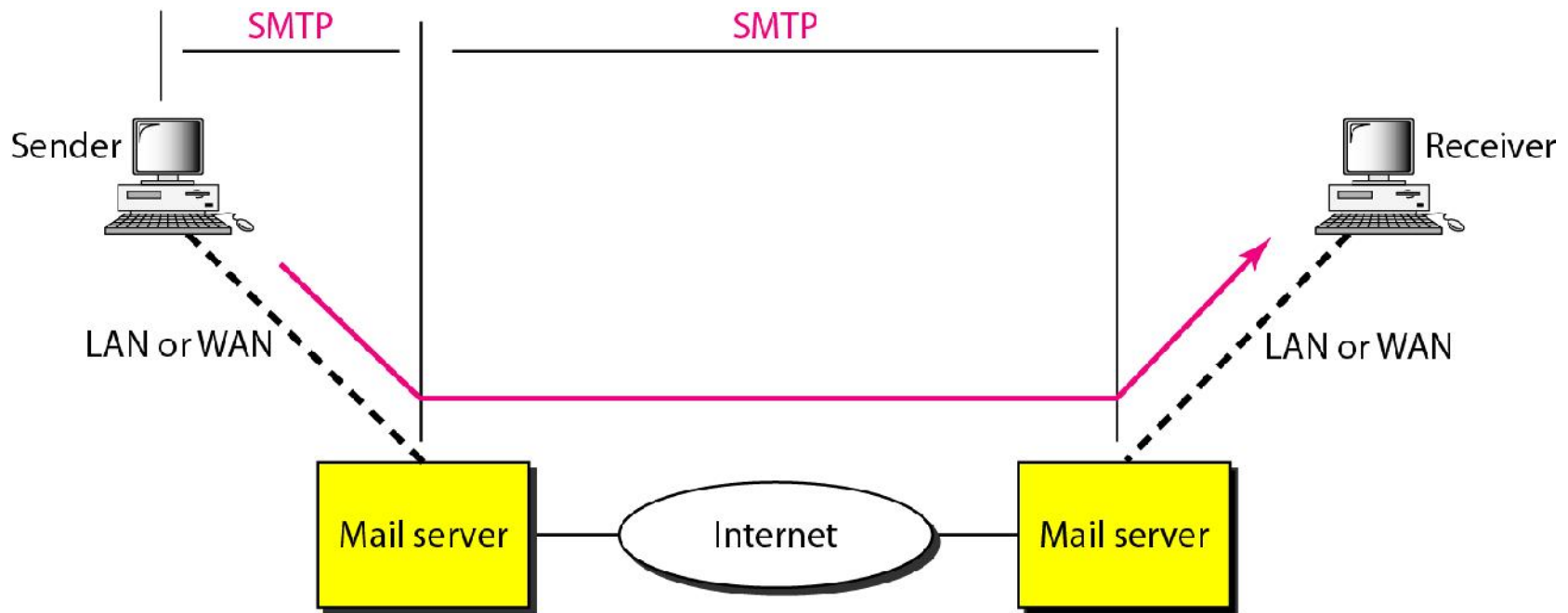
MIME header



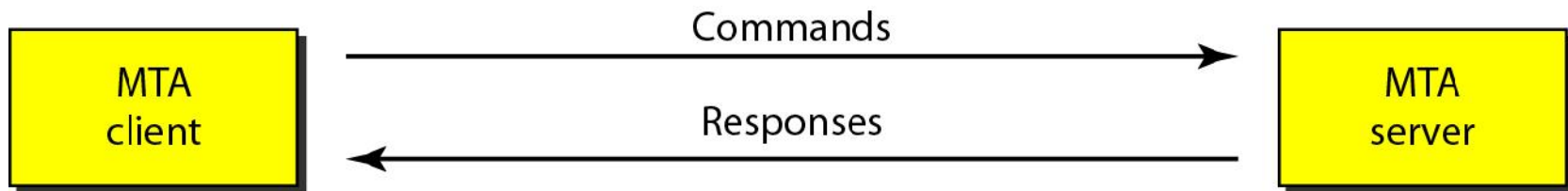
Data types and subtypes in MIME

<i>Type</i>	<i>Subtype</i>	<i>Description</i>
Text	Plain	Unformatted
	HTML	HTML format (see Chapter 27)
Multipart	Mixed	Body contains ordered parts of different data types
	Parallel	Same as above, but no order
	Digest	Similar to mixed subtypes, but the default is message/RFC822
	Alternative	Parts are different versions of the same message
Message	RFC822	Body is an encapsulated message
	Partial	Body is a fragment of a bigger message
	External-Body	Body is a reference to another message
Image	JPEG	Image is in JPEG format
	GIF	Image is in GIF format
Video	MPEG	Video is in MPEG format
Audio	Basic	Single-channel encoding of voice at 8 kHz
Application	PostScript	Adobe PostScript
	Octet-stream	General binary data (8-bit bytes)

Simple Mail Transfer Agent - SMTP



Commands and responses



Keyword: argument(s)

Commands

<i>Keyword</i>	<i>Argument(s)</i>
HELO	Sender's host name
MAIL FROM	Sender of the message
RCPT TO	Intended recipient of the message
DATA	Body of the mail
QUIT	
RSET	
VERFY	Name of recipient to be verified
NOOP	
TURN	
EXPN	Mailing list to be expanded
HELP	Command name
SEND FROM	Intended recipient of the message
SMOL FROM	Intended recipient of the message
SMAL FROM	Intended recipient of the message

Responses

<i>Code</i>	<i>Description</i>
Positive Completion Reply	
211	System status or help reply
214	Help message
220	Service ready
221	Service closing transmission channel
250	Request command completed
251	User not local; the message will be forwarded
Positive Intermediate Reply	
354	Start mail input
Transient Negative Completion Reply	
421	Service not available
450	Mailbox not available
451	Command aborted: local error
452	Command aborted: insufficient storage

Responses Cont...

<i>Code</i>	<i>Description</i>
Permanent Negative Completion Reply	
500	Syntax error; unrecognized command
501	Syntax error in parameters or arguments
502	Command not implemented
503	Bad sequence of commands
504	Command temporarily not implemented
550	Command is not executed; mailbox unavailable
551	User not local
552	Requested action aborted; exceeded storage location
553	Requested action not taken; mailbox name not allowed
554	Transaction failed

Example

```
===== Mail Transfer =====  
MAIL FROM: forouzanb@adelphia.net  
    250 Sender <forouzanb@adelphia.net> Ok  
RCPT TO: forouzanb@adelphia.net  
    250 Recipient <forouzanb@adelphia.net> Ok  
DATA  
    354 Ok Send data ending with <CRLF>.<CRLF>  
From: Forouzan  
TO: Forouzan  
  
This is a test message  
to show SMTP in action.  
•
```

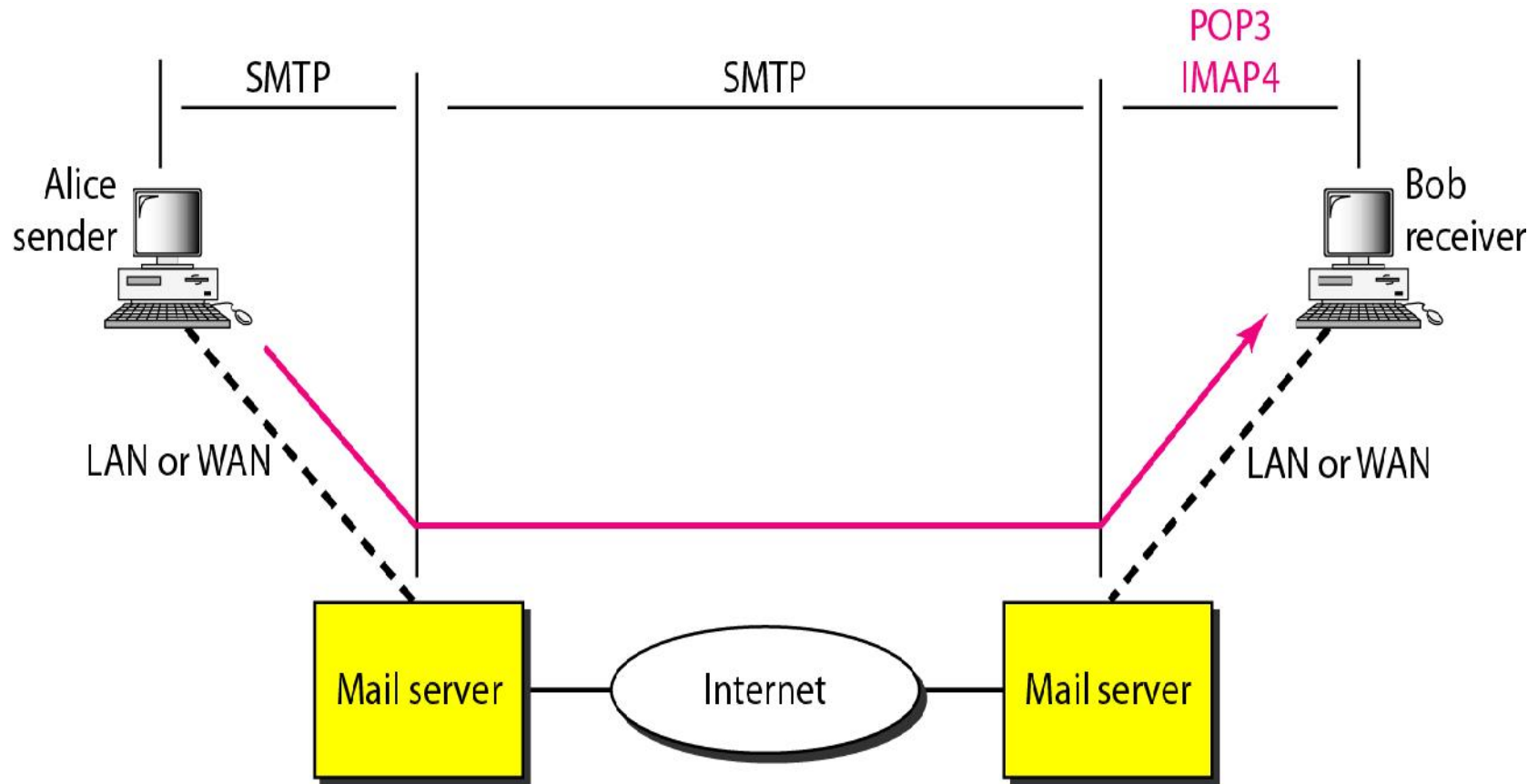
Example

```
===== Connection Termination =====  
250 Message received: adelphia.net@mail.adelphia.net  
QUIT  
221 mta13.adelphia.net SMTP server closing connection  
Connection closed by foreign host.
```

Message Access Agent

- POP3 → Post Office Protocol Version3
- IMAP4 → Internet Mail Access Protocol Version4

POP3 and IMAP4



Commands and responses in POP3

